

ORIGINAL RESEARCH ARTICLE

Nicotinamide Riboside Preserves Cardiac Function in a Mouse Model of Dilated Cardiomyopathy

Editorial, see p 2274

BACKGROUND: Myocardial metabolic impairment is a major feature in chronic heart failure. As the major coenzyme in fuel oxidation and oxidative phosphorylation and a substrate for enzymes signaling energy stress and oxidative stress response, nicotinamide adenine dinucleotide (NAD⁺) is emerging as a metabolic target in a number of diseases including heart failure. Little is known on the mechanisms regulating homeostasis of NAD⁺ in the failing heart.

METHODS: To explore possible alterations of NAD⁺ homeostasis in the failing heart, we quantified the expression of NAD⁺ biosynthetic enzymes in the human failing heart and in the heart of a mouse model of dilated cardiomyopathy (DCM) triggered by Serum Response Factor transcription factor depletion in the heart (SRF^{HKO}) or of cardiac hypertrophy triggered by transverse aorta constriction. We studied the impact of NAD⁺ precursor supplementation on cardiac function in both mouse models.

RESULTS: We observed a 30% loss in levels of NAD⁺ in the murine failing heart of both DCM and transverse aorta constriction mice that was accompanied by a decrease in expression of the nicotinamide phosphoribosyltransferase enzyme that recycles the nicotinamide precursor, whereas the nicotinamide riboside kinase 2 (NMRK2) that phosphorylates the nicotinamide riboside precursor is increased, to a higher level in the DCM (40-fold) than in transverse aorta constriction (4-fold). This shift was also observed in human failing heart biopsies in comparison with nonfailing controls. We show that the *Nmrk2* gene is an AMP-activated protein kinase and peroxisome proliferator-activated receptor α responsive gene that is activated by energy stress and NAD⁺ depletion in isolated rat cardiomyocytes. Nicotinamide riboside efficiently rescues NAD⁺ synthesis in response to FK866-mediated inhibition of nicotinamide phosphoribosyltransferase and stimulates glycolysis in cardiomyocytes. Accordingly, we show that nicotinamide riboside supplementation in food attenuates the development of heart failure in mice, more robustly in DCM, and partially after transverse aorta constriction, by stabilizing myocardial NAD⁺ levels in the failing heart. Nicotinamide riboside treatment also robustly increases the myocardial levels of 3 metabolites, nicotinic acid adenine dinucleotide, methylnicotinamide, and N1-methyl-4-pyridone-5-carboxamide, that can be used as validation biomarkers for the treatment.

CONCLUSIONS: The data show that nicotinamide riboside, the most energy-efficient among NAD precursors, could be useful for treatment of heart failure, notably in the context of DCM, a disease with few therapeutic options.

Nicolas Diguët, PhD
Samuel A.J. Trammell, PhD
Cynthia Tannous, PhD*
Robin Deloux, MSc*
Jérôme Piquereau, PhD
Nathalie Mougenot, PhD
Anne Gouge, MSc
Mélanie Gressette, MSc
Boris Manoury, PhD
Jocelyne Blanc, MSc
Marie Breton, MD
Jean-François Decaux, PhD
Gareth G. Lavery, PhD
István Baczkó, MD, PhD
Joffrey Zoll, PhD
Anne Garnier, PhD
Zhenlin Li, PhD
Charles Brenner, PhD
Mathias Mericskay, PhD

*Drs Tannous and Deloux contributed equally.

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Clinical Perspective

What Is New?

- Myocardial nicotinamide adenine dinucleotide (NAD) coenzyme levels are depressed in mouse models of heart failure.
- An expression shift occurs in murine and human failing hearts in which the normally predominant nicotinamide phosphoribosyltransferase enzyme using nicotinamide as a precursor for NAD⁺ synthesis is repressed, whereas the nicotinamide riboside kinase 2 enzyme using the nucleoside nicotinamide riboside (NR) is strongly upregulated.
- The nicotinamide riboside kinase 2 pathway is activated by energy stress in an AMP-activated protein kinase α /peroxisome proliferator-activated receptor α -dependent manner in cardiomyocytes.
- NR-supplemented diet administered to murine models of dilated cardiomyopathy or pressure overload-induced heart failure restores myocardial NAD levels and preserves cardiac function.
- NR increases glycolysis, and citrate and acetyl-coenzyme A metabolism, as well, in cardiomyocytes.

What Are the Clinical Implications?

- NR is the most energetically favorable NAD precursor and was recently shown to be uniquely and orally bioavailable in humans in a phase I clinical trial.
- NR-supplemented diet could help patients who have heart failure to cope with limited myocardial ATP supply by restoring NAD coenzyme levels and associated signaling.
- NR-supplemented diet has been shown in several studies to enhance metabolic flexibility in the context of obesity and diabetes mellitus and to improve muscle function, justifying that NR is put in the front line for metabolic therapy of heart failure syndrome in which these functions are altered.

Despite progress in therapeutic advances, mortality remains high in congestive heart failure (HF), creating a need for new evidence-based and cost-effective treatments. Failure of cardiomyocyte bioenergetics is a key issue in HF that is not well addressed in current therapeutics.^{1,2} We focused our attention on pathways that regulate nicotinamide adenine dinucleotide (NAD⁺) homeostasis in the failing heart, which has emerged as a new avenue for the development of metabolic therapy of HF.^{3,4} NAD⁺ is the major hydride transfer coenzyme in fuel oxidation and mitochondrial ATP generation and is the precursor for NADPH, which is required for anabolic pathways and reactive oxygen species detoxification.^{5,6} In addition, NAD⁺ is an essential substrate of enzymes including sirtuins and poly(ADP-ribose) polymerases, which are important in responding to altered nutritional

status and genotoxic stress.^{6,7} Sirtuins perform NAD⁺-dependent deacylation of protein Lys residues modified by acetyl coenzyme A (Ac-CoA) derived acyl groups.⁸ Sirtuins regulate nuclear gene expression, and mitochondrial enzyme activities, as well, and can play a protective role in the heart,⁹ although chronic overactivation may be deleterious.¹⁰ Poly(ADP-ribose) polymerase 1 is activated by reactive oxygen species-induced DNA lesions, and could deplete myocardial NAD⁺ stores in HF.¹¹ In situations of physiological equilibrium, biosynthetic pathways relying on dietary sources of tryptophan and NAD⁺ precursor vitamins counterbalance the consumption of NAD⁺ by signaling enzymes. However, dietary precursors may become insufficient to maintain the NAD⁺ metabolome in pathological conditions, thereby necessitating supplementation of NAD⁺ precursor.^{5,12,13}

The homeostasis of NAD⁺ and regulation of NAD⁺ biosynthetic enzymes have not been studied in dilated cardiomyopathy (DCM), a leading cause of heart transplant.¹⁴ We previously generated a powerful model of nonischemic DCM in mice triggered by cardiac-specific inducible inactivation of Serum Response Factor (*SRF^{HKO}* model).^{15–18} SRF is a major transcriptional regulator of genes and microRNAs involved in contractility, and in energy flux and calcium handling, as well.^{15,17,18} The SRF pathway is altered in the human failing heart^{19,20} and in animal models of DCM.^{21,22} The heart of *SRF^{HKO}* mutant mice display an early downregulation of myofibrillar creatine kinase (CK) activity encoded by the *Ckm* gene, which is followed by repression of enzymes of fatty acid β -oxidation and loss of ATP, thus recapitulating the transitional steps toward energy failure and cardiac structural remodeling in HF.¹⁸

Here, we show that NAD⁺ homeostasis is altered at an early stage in the heart of *SRF^{HKO}* mice with a striking induction of the nicotinamide riboside kinase 2 (*NMRK2*) pathway for NAD⁺ synthesis. *Nmrk2* gene is 1 of 2 mammalian paralogs of the yeast *NRK1* gene that was discovered as a eukaryotic route to NAD⁺ from nicotinamide riboside (NR).²³ Given the activity of NR in promoting sirtuin activity and restoring metabolic balance in a variety of model organisms,^{13,24–28} we aimed to test whether NR might treat HF in *SRF^{HKO}* mice and in the pressure overload hypertrophy model induced by transverse aorta constriction (TAC). Here, we show that NR greatly protects cardiac function by virtue of improved citrate and Ac-CoA metabolism and antioxidant gene expression. Because NR is orally available in people,²⁹ our data rationalize clinical testing of NR in human HF.

METHODS

The data, analytic methods, and study materials will be made available from the corresponding author on request to other researchers for purposes of reproducing the results or replicating the procedure. Detailed methods are available in the [online-only Data Supplement](#).

Patients

Left ventricular (LV) myocardium was obtained from terminally failing human hearts of 4 patients (mean age 54 years $\pm$ 7, SD) at the time of transplantation at the Hôpitaux Universitaires de Strasbourg as previously published with approval of the Hôpitaux Universitaires de Strasbourg ethics committee.³⁰ All patients gave informed consent. Patients' characteristics are detailed in [Table I in the online-only Data Supplement](#). The investigations conformed to the principles of the Declaration of Helsinki. Control hearts were obtained from the department of Pharmacology and Pharmacotherapy, University of Szeged, Szeged, Hungary. Experimental protocols were approved by the Ethical Review Board of the Medical Center of the University of Szeged and by the Scientific and Research Ethical Committee of the Medical Scientific Board at the Hungarian Ministry of Health (ETT-TUKEB; No. 51–57/1997 OEj and 4991-0/2010-1018EKU).

Transgenic Mice

All experiments with animals conformed to the Directive 2010/63/EU of the European Parliament and were approved by the ethics committee Charles Darwin #5 (agreement 00369.01). See [online-only Data Supplement](#) for details on SRF inactivation protocol.

NR Supplementation

SRF^{flKO} and control (*Sf/Sf*) mice were administered A04 rodent maintenance diet (Scientific Animal Food Engineering) supplemented with NR chloride provided by Chromadex. See [online-only Data Supplement](#) for details.

Transcriptomic Analysis

Transcriptomic data are available on NCBI Gene Expression Database, Accession number GSE84142.

Statistical Analysis

Animals were randomly assigned into different treatment groups. To assess significance, we performed Student *t* test for independent samples when the experimental design compared only 2 groups, or 1-way ANOVA for multiple group comparisons, or 2-way factorial ANOVA for independent factors when appropriate. Tukey tests were used for comparison of specific experimental groups when 1-way ANOVA gave a *P* value of <0.05 or, in the case of 2-way ANOVA for interaction, a *P* value of <0.05. Values are expressed as mean $\pm$ SEM.

Supplemental Information

Supplemental information includes supplemental methods, 9 supplemental figures, and 6 supplemental tables.

RESULTS

NMRK2 Pathway Is Activated in the Failing Heart of *SRF^{flKO}* Mice

Transcriptome analyses in *SRF^{flKO}* mice at baseline or after phenylephrine α -adrenergic stimulation¹⁶ revealed

that the *Nmrk2* gene was induced at an early stage in the *SRF^{flKO}* heart and rose continuously during establishment of HF (Figure 1A and [Table II in the online-only Data Supplement](#)). The transcript for the ectoenzyme NT5E (CD73) that hydrolyzes extracellular NAD⁺ and nicotinamide mononucleotide (NMN) to NR, the substrate of NMRK2 kinase³¹ also increased in *SRF^{flKO}* hearts ([Table II in the online-only Data Supplement](#)). In contrast, the transcripts encoding nicotinamide phosphoribosyltransferase (NAMPT), the enzyme converting nicotinamide (NAM) to NMN, and purine nucleoside phosphorylase (PNP), which converts NR to NAM,³² were depressed on phenylephrine treatment in *SRF^{flKO}* hearts ([Table II in the online-only Data Supplement](#)). Genes involved in the tryptophan and nicotinic acid pathways for NAD⁺ synthesis were not altered (Figure 1A and [Table II in the online-only Data Supplement](#)).

Modulation of *Nmrk2* and *Nampt* gene expression detected by transcriptomics was corroborated at the mRNA and protein level (Figure 1B through 1D). We observed a similar shift in human patients from predominant expression of NAMPT in healthy hearts to increased NMRK2 and depressed NAMPT in failing hearts (Figure 1D, [Table II in the online-only Data Supplement](#)). In mouse heart sections, NMRK2 protein localizes at the proximity of sarcolemma in controls (Figure 1E). The signal was strongly enhanced and spread to cytoplasm in *SRF^{flKO}* cardiomyocytes.

At day 15 after SRF inactivation, there was a 30% loss of NAD (representing the sum of NAD⁺ and NADH) (Figure 1F, see [Methods in the online-only Data Supplement](#) for details). The gene expression pattern (*Nt5e* and *Nmrk2* up with *Pnp* and *Nampt* down) suggested that cardiac tissue is attempting to mobilize and utilize NR as an NAD⁺ precursor while not increasing NAM usage. We tested this hypothesis by intraperitoneal administration of NR and NAM to *SRF^{flKO}* mice from day 8 to 15. Consistent with this hypothesis, myocardial NAD levels were preserved by NR but not by NAM (Figure 1F). *Srf* and *Nmrk2* expression levels were not changed by these treatments (Figure 1G and 1H). Because NR is orally available,^{26,29} we tested an NR-supplemented diet for 2 weeks. Myocardial NAD levels were reduced in this second series of *SRF^{flKO}* mice in comparison with controls and rescued by NR administration (Figure 1I).

NR Preserves Cardiac Function in the *SRF^{flKO}* Heart

We hypothesized that NR supplementation of food might be beneficial for cardiac function in the context of DCM. We fed control and mutant *SRF^{flKO}* mice a standard chow diet or NR-supplemented diet to reach a dose of 400 mg/kg of body weight per day from day 5 to 50, a period in which untreated *SRF^{flKO}* mice develop DCM and progress toward HF.^{15–18} NR diet induced a

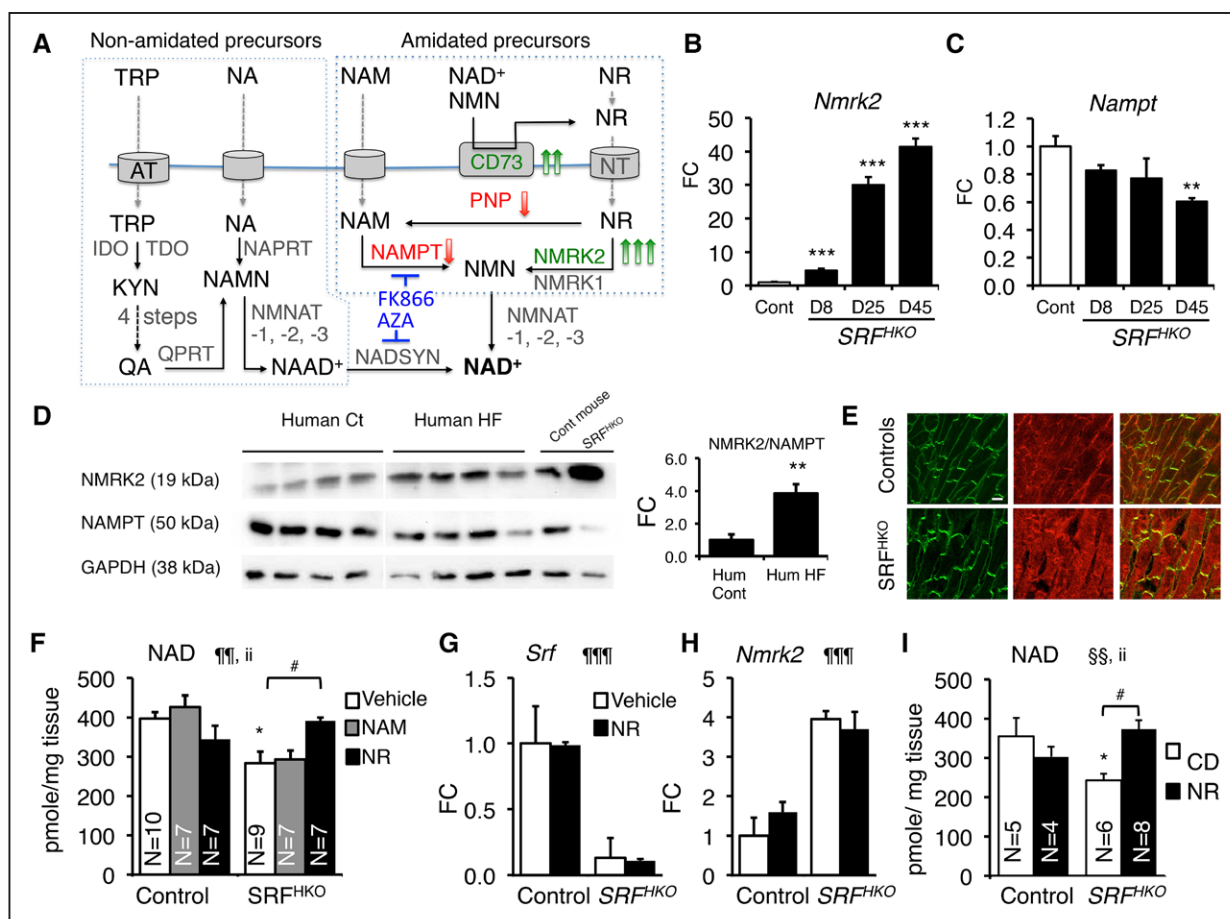


Figure 1. Activation of the *Nmrk2* gene in dilated cardiomyopathy.

A, Biosynthetic NAD⁺ pathways in mammalian cells. NA (nicotinic acid), NAM, NR, and TRP enter cells through specialized transport systems. Extracellular NAD⁺ and NMN are hydrolyzed to NR by the ectonucleotidase CD73. In the de novo pathway, TRP is converted to kynurenin (KYN) by IDO and TDO enzymes, and then into quinolinic acid (QA) in 4 steps (not detailed). QA, NA, and NAM are phosphoribosylated by QPRT, NAPRT1, and NAMPT enzymes, respectively, to form mononucleotides NAMN and NMN. NR is phosphorylated by NMRK enzymes to form NMN. NR can be converted to NAM by PNP. NAMN and NMN are adenylated by the NMNAT enzymes to form the dinucleotides NAAD⁺ and NAD⁺. NAAD⁺ is amidated by NADSYN1. Genes increased in *SRF^{HKO}* heart are highlighted in green and those repressed in red (see accompanying Table II in the online-only Data Supplement). Inhibitors are shown in blue. **B** and **C**, RT qPCR analysis of *Nmrk2* and *Nampt* mRNA fold change (FC) in *SRF^{HKO}* hearts at different days (D) after SRF inactivation. **D**, Western blot analysis of NMRK2 and NAMPT protein in control and failing human hearts and control and *SRF^{HKO}* mouse hearts at D45. GAPDH is used as loading control. All human and mouse samples were analyzed on single blot, but 2 lanes separating that showed signs of protein degradation and aberrant migration pattern were cut out of the picture. **Right**, Ratio of NMRK2/NAMPT in human samples. **E**, Control and *SRF^{HKO}* mutant heart sections stained for NMRK2 (red) and vinculin (green) at D45. White bar in upper left panel = 20 μm. **F** through **H**, NAM or NR (30 μmol) were injected intraperitoneally from D8 to 15 (400 mg/kg per day) to control and *SRF^{HKO}* mice. Vehicle: saline solution. **F**, Myocardial NAD⁺ levels. **G** and **H**, RT qPCR analysis of *Srf* and *Nmrk2* mRNA levels. **I**, Myocardial NAD levels in control and *SRF^{HKO}* mice fed regular chow diet (CD) or NR-enriched diet (0.22%, 400–450 mg/kg per day) from D5 to 20. Throughout the figure, data are expressed as mean ± SEM. Statistical analysis: One-way ANOVA (**B**, **C**) or 2-way factorial ANOVA for independent samples (**F** through **I**). ¶¶¶*P* ≤ 0.01, ¶¶¶¶*P* ≤ 0.001 for the genotype effect; §§*P* ≤ 0.01 for the NR treatment effect; ii *P* ≤ 0.01 for the interaction effect. Post hoc Tukey test: asterisks indicate statistical significant difference between any group versus the control CD (or vehicle) group: **P* ≤ 0.05. ***P* ≤ 0.01. ****P* ≤ 0.001. # *P* ≤ 0.05 for the effect of NR within the *SRF^{HKO}* group. A *t* test was used for the graph at the right of **D**: ** *P* ≤ 0.01. FC indicates fold change over control group; IDO, indoleamine 2,3-dioxygenase; NAAD, nicotinic acid adenine dinucleotide; NAD, nicotinamide adenine dinucleotide; NAM, nicotinamide; NAMN, nicotinic acid mononucleotide; NAMPT, nicotinamide phosphoribosyltransferase; NMN, nicotinamide mononucleotide; NMNAT, nicotinamide mononucleotide adenyltransferase; NMRK, nicotinamide riboside kinase; NR, nicotinamide riboside; PNP, purine nucleoside phosphorylase; QPRT, quinolinic acid phosphoribosyltransferase; RT qPCR, quantitative reverse transcription polymerase chain reaction; SRF, Serum Response Factor; TDO, tryptophan 2,3-dioxygenase; and TRP, tryptophan.

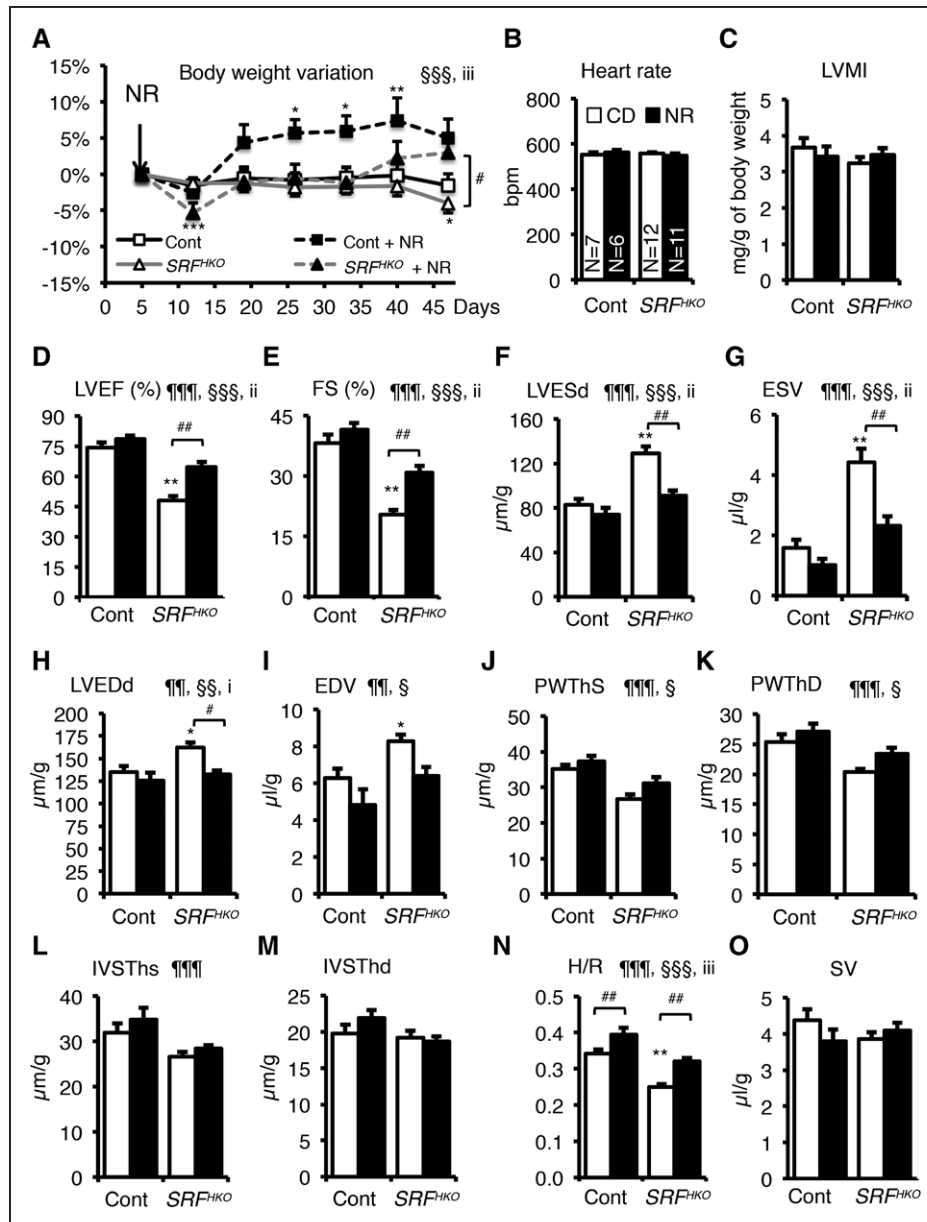


Figure 2. NR supplementation in diet prevents the onset of heart failure and dilatation.

A, CD- or NR-supplemented diet (0.22%) was given ad libitum to control and *SRF^{HKO}* mice from day 5 after SRF inactivation to the end of the experiment. Body weight was monitored throughout the period. Data are expressed as mean % weight variation $\pm$ SEM, in comparison with weight at day 5. A 2-factor ANOVA with repeated measures on 1 factor was used for statistical analysis. §§§ $P\leq 0.001$ for the time effect; iii, $P\leq 0.001$ for the interaction effect. Post hoc Tukey test: * $P\leq 0.05$, ** $P\leq 0.01$, *** $P\leq 0.001$ for any time point versus day 5 within a group; # $P\leq 0.05$ for the NR treatment in *SRF^{HKO}* group. **B** through **O**, Cardiac parameters of control and *SRF^{HKO}* mutant were analyzed in M-Mode echocardiography between day 45 and day 47. **B**, Heart rate; **C**, left ventricle (LV) mass index; **D**, LV ejection fraction; **E**, fractional shortening; **F** and **G**, LV end-systolic diameter (**F**) and volume (**G**); **H** and **I**, LV end-diastolic diameter (**H**) and volume (**I**); **J** and **K**, LV posterior wall thickness in systole (**J**) and diastole (**K**); **L** and **M** interventricular septum thickness in systole (**L**) and diastole (**M**); **N**, H/R: LV thickness (H) to radius (R) ratio; and **O**, stroke volume. Dimensions were normalized by the body weight. Data are expressed as mean $\pm$ SEM. Statistical analysis: 2-way factorial ANOVA for independent samples. §§§ $P\leq 0.001$ for the genotype effect; § $P\leq 0.05$, §§ $P\leq 0.01$, §§§ $P\leq 0.001$ for the NR treatment effect; i $P\leq 0.05$, ii $P\leq 0.01$, iii $P\leq 0.001$ for the interaction effect. Post hoc Tukey test: asterisks indicate statistically significant difference versus the control CD group: * $P\leq 0.05$, ** $P\leq 0.01$, *** $P\leq 0.001$. # $P\leq 0.05$, ## $P\leq 0.01$ for the effect of NR within the *SRF^{HKO}* group. CD indicates chow diet; CONT, control; EDV, end-diastolic volume; ESV, end-systolic volume; FS, fractional shortening; IVSThd, interventricular septum thickness in diastole; IVSThs, interventricular septum thickness in systole; LVEF, left ventricular ejection fraction; LVEDd, left ventricular end-diastolic diameter; LVESd, left ventricular end-systolic diameter; LVMI, left ventricular mass index; NR, nicotinamide riboside; PWThd, posterior wall thickness in diastole; PWThs, posterior wall thickness in systole; and SV, stroke volume.

modest 5% to 7% increase in body weight in control mice and allowed *SRF^{FKO}* mice to regain weight after an initial loss, whereas *SRF^{FKO}* mice on a regular diet had lost weight at the end of the protocol (Figure 2A). Cardiac parameters were analyzed by echocardiography between days 45 and 47 (Figure 2B through 2O). NR did not change heart rate and LV mass index, but slightly it increased the LV thickness-to-radius ratio (H/R) in control mice (Figure 2N). To assess a potential impact of NR treatment on vascular function, we compared vascular reactivity in isolated mesenteric arteries from NR-fed and control mice. We found no difference between the 2 groups neither in responses to K⁺ or agonist-induced contraction nor in the relaxation response to carbachol or a NO donor (Figure I in the online-only Data Supplement and Table III in the online-only Data Supplement).

SRF^{FKO} mutant mice fed the standard diet displayed a severe decrease in LV ejection fraction and fractional shortening (Figure 2D and 2E). The NR diet clearly protected against this decline in cardiac function. It is important to note that NR fully protected *SRF^{FKO}* mice against the dilatation and thinning of the LV wall (Figure 2F through 2M), preserving the H/R ratio that normally declines with DCM (Figure 2N). Changes in stroke volume (Figure 2O) and cardiac output (not shown) were not significant.

At the histological level, NR limited the appearance of very long cardiomyocytes (Feret > 100 µm) in *SRF^{FKO}* mice, although the population of large cardiomyocytes (minFeret > 26 µm) remained larger than in controls (Figure II in the online-only Data Supplement).

NR-Enriched Diet Protects the Cardiac NAD⁺ Metabolome in HF

The impact of an NR-enriched diet on the cardiac NAD⁺ metabolome is not known (Figure 3A). Mice fed with or without NR were euthanized 3 days after the echocardiography at day 50. Myocardial NAD was decreased in the *SRF^{FKO}* mice at day 50 as measured by the NAD cycling assay, and the NR diet protected against this drop (Figure 3B). Myocardial NAD⁺ and NADH varied in the same direction in all conditions, leaving the NAD⁺/NADH ratio unchanged (Figure III in the online-only Data Supplement). Liquid chromatography-mass spectrometry analysis allowed us to provide the levels of NAD⁺ metabolites on a common scale with NAD⁺^{13,33} (Figure 3C through 3L). *SRF^{FKO}* mice hearts did not have higher levels of adenosine diphosphate ribose (ADPR) or NAM (Figure 3D and 3E). Levels of NADP⁺ and NMN showed a pattern similar to NAD⁺ (Figure 3F and 3G).

NR increased the cardiac levels of nicotinic acid adenine dinucleotide, a sensitive biomarker of increased NAD⁺ metabolism,²⁹ and methyl-NAM (MeNAM), as well (Figure 3A, 3H through 3J). MeNAM oxidation by AOX1 generates N1-methyl-4-pyridone-5-carboxamide (Me4PY)

and releases hydrogen peroxide (Figure 3A). Changes in unrelated metabolites were not significant (Figure 3K and 3L). As shown in the nematode,³⁴ NR increased the expression of the *Nfe12* gene encoding NRF2, whereas its target genes *Mt2* and *Nox4* were already activated, and *G6pdx*, as well, in the heart of *SRF^{FKO}* mice (Figure 3M through 3P). NR treatment in the *SRF^{FKO}* hearts limited the level of *Myh7* induction, a signature of cardiac stress and metabolic remodeling in the failing heart (Figure 3Q).

The liver is a central organ for the regulation of nutrient metabolism. We assessed the NAD⁺ metabolome response in the liver of control and *SRF^{FKO}* mice (Table IV in the online-only Data Supplement). NR increased the MeNAM and Me4PY levels similarly in both genotypes. It is interesting to note that the liver response to NR diet was more robust in the *SRF^{FKO}* mice with regard to the increase in the steady-state level of NAD, nicotinic acid adenine dinucleotide, ADPR, and NAM. Because the liver is not targeted by the *Srf* deletion, it suggests that the HF condition indirectly modifies the balance of NAD synthesis and signaling in the liver.

NR-Enriched Diet Does Not Result in Global Cardiac Protein Deacetylation

The cardiac levels of NAM and ADPR in this HF model are low and show few changes with NR (Figure 3D and 3E), suggesting that the NR-driven boost in NAD⁺ synthesis is not necessarily translated into a higher level of NAD⁺ consumption by sirtuins and poly(ADP-ribose) polymerases. We performed antiacetyl-K and anti-poly-ADP-ribose Western blot analyses on cardiac proteins from control and *SRF^{FKO}* mice fed with the chow diet or NR-enriched diets. Just as levels of ADPR were not modulated by *SRF^{FKO}* or NR, neither were levels of PAR (Figure IVA in the online-only Data Supplement). There was an increase in the acetylation level of some cardiac proteins in the hearts of NR-fed animals (Figure IVB in the online-only Data Supplement). Antibodies specific to the acetylated form of FOXO1 or p53 showed an increase in the acetylation level of these nuclear proteins in the heart in response to NR in controls as in *SRF^{FKO}* mice (Figure 3R, Figure V in the online-only Data Supplement). By comparison, the acetylation level of the mitochondrial aconitase 2 protein did not change in any of these conditions despite lower protein level in the *SRF^{FKO}* hearts (Figure 3S and Figure VIA in the online-only Data Supplement). These data suggest that the bioenergetic improvement in NR-supplemented hearts may result in greater production of cytosolic Ac-CoA and consequent increased nucleocytosolic protein acetylation.

NR Improves Metabolism of Citrate in HF

Several regulators involved in mitochondrial biogenesis and oxidative metabolism, and lipid metabolism, as

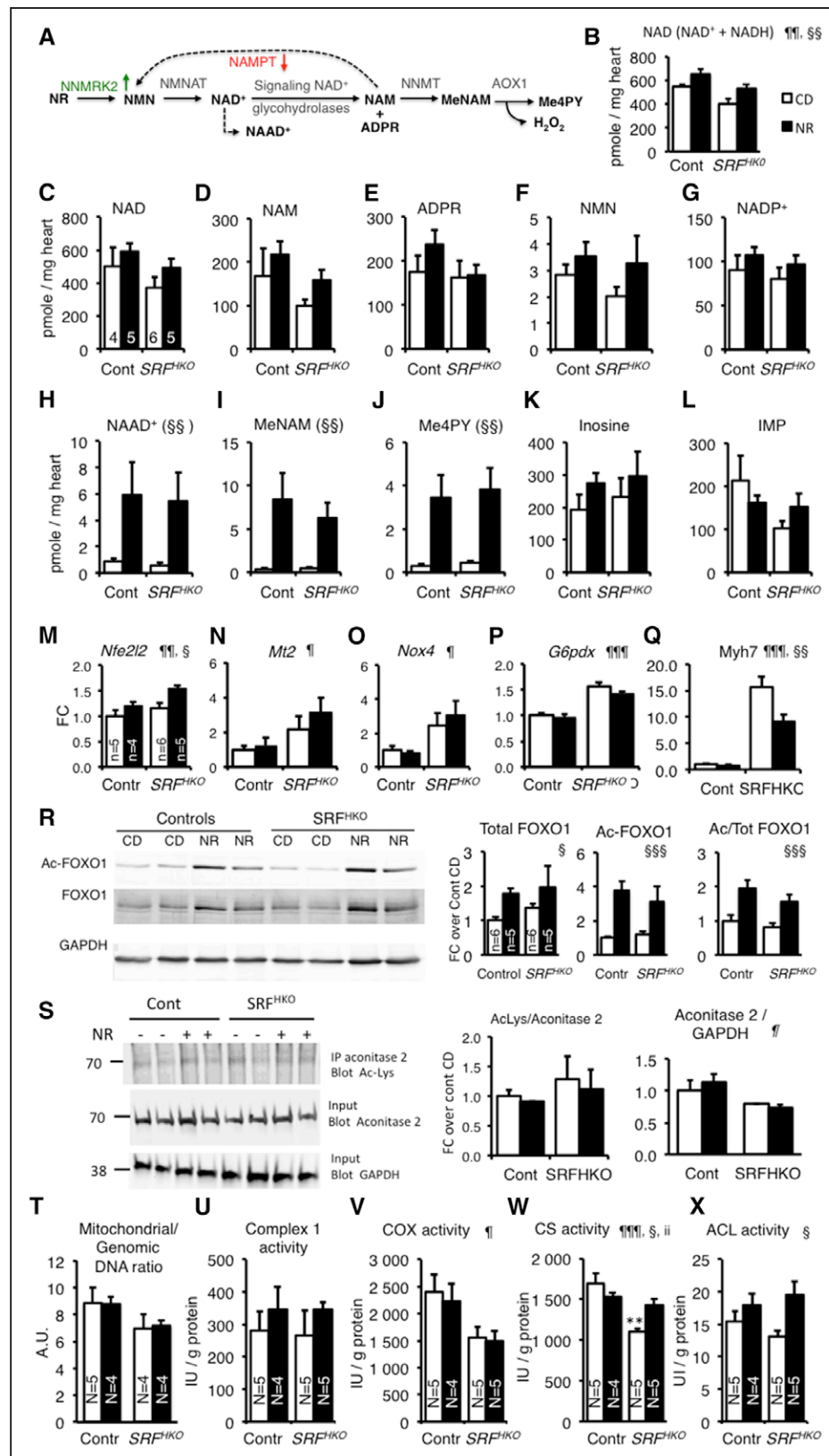


Figure 3. Impact of NR treatment on cardiac NAD⁺ metabolome.

A, NR to NAD⁺ pathway and NAD⁺ catabolism are depicted. NAD⁺ is cleaved into NAM and ADP-ribose by NAD⁺-consuming enzymes. NAM is either recycled by NAMPT or methylated by the nicotinamide *N*-methyltransferase (NNMT) enzyme and oxidized by the aldehyde deoxidase 1 (AOX1) giving rise to degradation products MeNAM and Me4PY, respectively. **B**, Myocardial NAD was quantified by colorimetric NAD⁺ cycling assay at day 50 in controls (n=6), Controls+NR (*Continued*)

well, were repressed in *SRF^{FKO}* hearts (Figure VIIA and VIIB in the online-only Data Supplement). Simultaneously, the expression of the glucose transporter *Glut1* was increased and expression of *Pdk2* was decreased, which can be seen as a trend of the failing heart to increase glycolysis (Figure VIIC in the online-only Data Supplement). NR did not alter the expression of these genes.

There was a trend toward a reduced mitochondrial to genomic DNA ratio in the *SRF^{FKO}* LV (Figure 3T). The NADH-dependent complex I activity and Cox activity was not changed by the NR diet (Figure 3U and 3V). Neither the complex I and II maximal respiration rates, nor the apparent K_m for ADP, mitochondrial CK coupling, and ADP acceptor control ratio showed any alteration at this stage (day 50) in the *SRF^{FKO}* LV fibers. These parameters were not altered by NR administration (Figure VIID through VIIG in the online-only Data Supplement).

We noted that NR treatment resulted in a net increase in acetylation of FOXO1 and p53 (Figure 3R and Figure V in the online-only Data Supplement). Availability of cytosolic Ac-CoA depends on mitochondrial citrate synthase (CS) production of citrate and conversion to Ac-CoA via the cytosolic enzyme, ATP-citrate lyase (ACL). CS activity was reduced to 65% of control levels in the *SRF^{FKO}* LV myocardium (Figure 3W). It is striking that NR administration protected against the decline of CS activity in the failing heart. This rescue in activity occurred despite a net reduction in protein level in the *SRF^{FKO}* myocardium that was not rescued by the NR treatment (Figure VIB in the online-only Data Supplement). It is interesting to that NR also increased cardiac ACL activity in both control and *SRF^{FKO}* mice (Figure 3X). Taken together, these data suggest that NR improves CS and ACL activity, resulting in increased nucleocytoplasmic protein acetylation.

Nmrk2 Gene Is Induced by Inhibition of Alternative NAD⁺ Biosynthetic Pathways in Cultured Cardiomyocytes

To assess the impact of NMRK2 on NAD⁺ biosynthesis relative to the other pathways in cardiomyocytes, we used chemical inhibitors to target the alternative pathways in neonatal rat cardiomyocytes (NRCs) grown in Dulbecco modified Eagle medium containing NAM as the vitamin B3. The glutamine analog azaserin inhibits the glutamine-dependent NAD synthase NADSYN1,³⁵ which catalyzes the last step of NAD⁺ synthesis from tryptophan and NA (Figure 1A). FK866 is a specific inhibitor of Nampt.³⁶

Azaserin had a slowly repressive effect on NAD pools that resulted in a 50% reduction with respect to nontreated cells at 72 hours (Figure 4A through 4C) while progressively increasing *Nmrk2* expression (Figure 4D). FK866 treatment led to a rapid and severe loss of intracellular NAD⁺ but not NADH (Figure 4A through 4C). Despite this severe loss, NRC could be maintained up to 72 hours in continuous layer with FK866 treatment. FK866 treatment strongly increased *Nmrk2* expression at 72 hours (Figure 4D). Combined azaserin and FK866 treatment had a similar impact than FK866 alone (Figure 4A through 4D). *Nmrk1* expression did not change in these conditions (not shown).

We tested whether NR could preserve NAD⁺ levels in the presence of FK866. NR slightly increased the baseline level of intracellular NAD⁺ in nontreated NRC, whereas exogenous NAD⁺ had no impact (Figure 4E). Both compounds increased the NAD⁺/NADH ratio (Figure 4F and 4G). NR fully protected NAD⁺ levels in NRCs treated with FK866, whereas the protection by exogenous NAD⁺ was partial (Figure 4E), although both treat-

Figure 3 Continued. (n=5), *SRF^{FKO}* (n=6), *SRF^{FKO}*+NR (n=6). Data are expressed as means±SEM. **C** through **L**, Myocardial metabolites were analyzed by LCMS-based metabolomics. See abbreviations in the text. Controls (n=4), controls+NR (n=5), *SRF^{FKO}* (n=6), +NR (n=5, except for NAM, MeNAM, and Me4PY, n=4, no peak was identified in 1 sample). Data are expressed as means±SEM. **M** through **Q**, Expression of genes related to oxidative stress signaling (**M** through **O**) and cardiac structural and metabolic remodeling (**P** and **Q**). n=5 in each group. **R**, Representative Western blot analysis of FOXO1 and acetyl-FOXO1 of 3 independent experiments realized in different duplicates for each group. GAPDH is used as a loading control. **Right graphs**, Quantification of total and acetyl-FOXO1 ratio on GAPDH and acetyl-FOXO1/total FOXO1 ratio in n=5 to 6 animals per group. **S**, Acetylation level of mitochondrial aconitase 2. Aconitase 2 (75 kDa) was immunoprecipitated using a rabbit polyclonal antibody, and the immunoprecipitate was analyzed by Western blot using a mouse monoclonal anti-Ac(K103) antibody. Inputs were run in a parallel gel and immunoblotted with anti-aconitase 2 and anti-GAPDH for loading control. The ratio of acetyl-aconitase2/total aconitase2 and aconitase2/GAPDH are shown (**right**). See accompanying Figure VIA in the online-only Data Supplement for control immunoprecipitation with preimmune rabbit IgG. **T** through **X**, LV cardiac tissue was isolated at 50 days after tamoxifen injection in control and *SRF^{FKO}* mice fed control diet (CD) or NR-enriched diet. DNA and proteins were extracted from parallel samples to quantify mitochondrial to genomic DNA ratio (**T**) and enzymatic activities: complex I (**U**), cytochrome oxidase (**V**), citrate synthase (**W**), and ATP citrate lyase (**X**). See associated Figure VII in the online-only Data Supplement. In **M** through **S**, data are expressed as mean fold change (FC)±SEM over control group CD. Statistical analysis: Two-way factorial ANOVA for independent samples was used for all panels. ¶*P*≤0.05, ¶¶*P*≤0.01, ¶¶¶*P*≤0.001 for the genotype effect; §*P*≤0.05, §§*P*≤0.01, §§§*P*≤0.001 for the NR treatment effect; ii *P*≤0.01 for the interaction effect. Post hoc Tukey test: asterisks indicate statistically significant difference versus the control CD group: ***P*≤0.01. ACL indicates ATP-citrate lyase; ADP, adenosine diphosphate ribose; Cont, control; CS, citrate synthase; IMP, inosine monophosphate; LCMS, liquid chromatography-mass spectrometry; NAAD, nicotinic acid adenine dinucleotide; NAD, nicotinamide adenine dinucleotide; NAM, nicotinamide; NAMPT, nicotinamide phosphoribosyltransferase; NMN, nicotinamide mononucleotide; and NR, nicotinamide riboside.

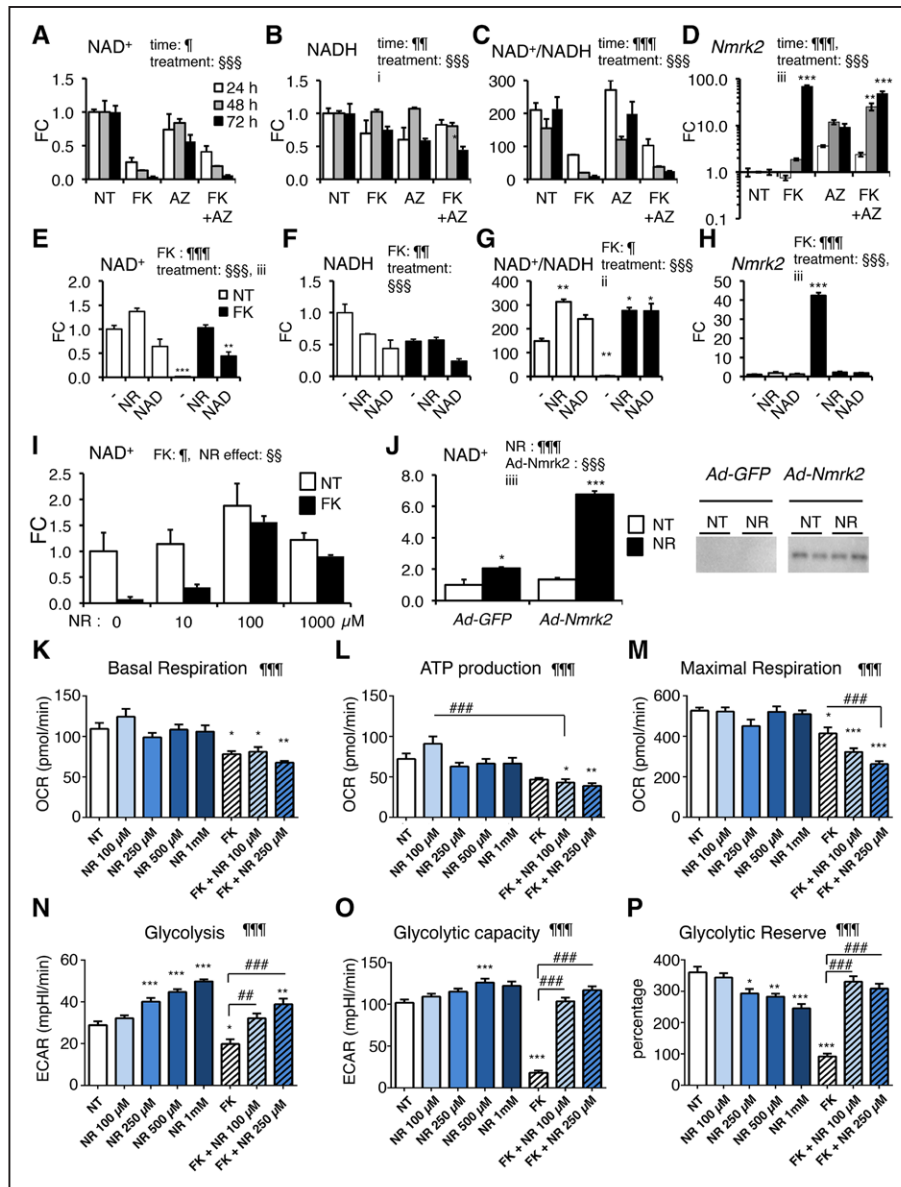


Figure 4. *Nmrk2* expression is activated by repression of alternative NAD⁺ biosynthetic pathways.

A through C, Intracellular levels of NAD⁺ (**A**), NADH (**B**), and NAD⁺/NADH ratio (**C**) in NRC after 10 μmol/L FK866 and 20 μmol/L azaserin treatment or no treatment (NT) for 24 to 72 hours as indicated. **D,** *Nmrk2* mRNA level in NRC treated as in **A** through **C**. **E through G,** Same as in **A** through **C** in NRC treated for 72 hours with 10 μmol/L FK866 or not treated (NT) in normal culture medium (–) or in presence of 250 μmol/L NAD⁺ or 1 mmol/L NR. **H,** *Nmrk2* mRNA level in NRC treated as in **E** through **G**. **I,** NAD⁺ levels in NRCs treated with 10 μmol/L FK866 for 72 hours, or not treated (NT) in the presence of increasing concentration of NR in culture medium. **J,** NAD⁺ content in nontreated (NT) or following 24-hour NR treatment (1 mmol/L) in NRC infected with Ad-GFP or HA-Nmrk2. **Right,** Western blot detection of HA-NMRK2 with anti-HA antibody. **K through M,** Mitochondrial stress test in Seahorse analyzer. NRC grown on Seahorse 96-well plates were analyzed for oxygen consumption rate (OCR) at day 8 after 5 days of treatment. **K,** Basal mitochondrial respiration is calculated from total cellular respiration minus nonmitochondrial respiration. **L,** ATP production is calculated from basal mitochondrial respiration minus respiration after oligomycin injection. **M,** Maximal respiration is measured after FCCP injection. See accompanying [Figure IX in the online-only Data Supplement](#) for other respiration parameters. **N through P,** Glycolysis stress test in Seahorse analyzer. NRC grown on Seahorse 96-well plates were analyzed at day 8 after 5 days of treatment. **N,** Glycolysis was measured as a function of extracellular acidification rate (ECAR) after injection of 10 mmol/L glucose. **O,** Glycolytic capacity as the maximum ECAR following injection of oligomycin. **P,** Glycolytic reserve as the difference between glycolysis and maximal glycolytic capacity. Throughout the figure, data are expressed as mean fold change (FC)±SEM over the control group, except when indicated. Statistical analysis: a 2-way factorial ANOVA for independent samples was used for **A** through **J**. ¶ and § symbols as indicated in the panels; i P≤0.05, ii P≤0.01, iii P≤0.001 for the interaction effect. One-way ANOVA was used for **K** through **P**. ¶ P<0.05, ¶¶ P<0.01, ¶¶¶ P<0.001. §§ P≤0.01, §§§ P≤0.001. Post hoc Tukey test: *P<0.05, **P<0.01, ***P<0.001 between any (Continued)

ments blocked *Nmrk2* induction (Figure 4E). At least 100 $\mu\text{mol/L}$ NR was required to rescue the NAD^+ loss induced by 10 $\mu\text{mol/L}$ FK866 (Figure 4I). To assess the functionality of NMRK2 in the absence of a stress like the FK866, we infected NRC with a recombinant adenovirus expressing HA-tagged *Nmrk2* cDNA (Figure 4J). Overexpression of NMRK2 did not modulate the NAD^+ levels in NRC. In presence of NR, *Nmrk2* overexpression robustly increased the NAD^+ level by a factor of 6. NR alone slightly increased the NAD^+ level in control NRC infected with *Ad-GFP* (Figure 4J). In isolated primary cultures of adult rat cardiomyocytes, NR increased intracellular NAD nearly 3-fold, showing that the NMRK pathway is more active in adult rat cardiomyocytes than in NRCs (Figure VIIIA in the online-only Data Supplement). FK866 reduced NAD levels by a factor of 2 (Figure VIIIA in the online-only Data Supplement) in adult rat cardiomyocytes and increased *Nmrk2* expression by a factor of 4.5, whereas azaserin had no impact (Figure VIIIB in the online-only Data Supplement). We assessed the impact of NR on oxygen consumption rates in NRC following a mitochondrial stress protocol. NR had no impact on any mitochondrial respiration parameter in NRC at baseline (Figure 4K through 4M, Figure IX in the online-only Data Supplement). NR increased glycolysis at baseline in a dose-dependent manner (Figure 4N), but not the maximal glycolytic capacity (Figure 4O and 4P). We assessed the capacity of NR to rescue any impact of FK866 on metabolism. FK866 reduced basal oxygen consumption rates and maximal respiration and trended to reduce ATP production (adjusted P value=0.0547) (Figure 4K through 4M). Addition of NR at doses that we showed to fully rescue the drop in NAD^+ levels triggered by the FK866 did not correct these alterations and even further reduced maximal respiration at high doses (Figure 4K through 4M, Figure IX in the online-only Data Supplement). FK866 strongly reduced basal glycolysis and glycolytic capacities in NRC. NR strikingly preserved NRC glycolytic capacities to 100% control levels and more in FK866-treated cells and stimulated glycolysis at baseline in nontreated cells in a dose-dependent manner (Figure 4N through 4P).

SRF Deficiency-Induced Depression of the *Nmrk2* Promoter Is Counterbalanced by the Sensitivity of *Nmrk2* Gene to NAD^+ Depletion and Energetic Stress

Because *Nmrk2* expression is activated in vivo on *Srf* gene inactivation, we addressed the role of the SRF transcription factor in *Nmrk2* gene regulation. We identified

an SRF binding site with 9/10 bp match to the consensus $\text{CC(A/T)}_6\text{GG}$ (CArG) sequence in a conserved 600-bp region between the murine and human genes (Figure 5A). We cloned the upstream region and 5'-UTR of the murine *Nmrk2* gene (−581/+61) into a luciferase reporter vector and mutated the CArG-like sequence to a motif matching the consensus sequence (5'-CC) or a motif unable to bind SRF (5'-GG) (Figure 5A). FK866 increased the expression of the WT construct 8-fold, whereas a consensus CArG reduced the extent of activation (Figure 5B). The 5'-GG mutation reduced baseline expression by 75% and abolished the response to FK866. The full intergenic 3009-bp sequence upstream of *Nmrk2* had a much higher activity, especially in response to FK866 (Figure 5C). siRNA-mediated inhibition of *Srf* reduced, but did not abolish, the induction of endogenous *Nmrk2* by FK866 in NRC, suggesting that FK866-triggered activation of the *Nmrk2* promoter is partially independent of SRF (Figure 5D and 5E). FK866 treatment reduces ATP to 50% of the control level in NRC (Figure 5F). SRF is major upstream regulator of *Ckm* gene involved in high-energy phosphate compounds balance. A siRNA-mediated 5-fold knock-down of *Ckm* gene expression lowered the cellular ATP levels to 70% control level and triggered a 6.8-fold increase in *Nmrk2* expression (Figure 5G and 5H). A 2-fold reduction in *Srf* expression resulted in a 2-fold reduction of *Ckm* expression leaving *Nmrk2* expression intact. Enforcing a more profound depression of *Ckm* expression in the context of *Srf* repression in NRC by cotransfection with both *Srf* and *Ckm* siRNAs resulted in a 2.3-fold increase in *Nmrk2* expression (Figure 5H).

Nmrk2 Is Induced by the AMPK-PPAR α Axis

In line with the previous results, we hypothesized that pathways related to energy failure might activate *Nmrk2*. The level of phosphorylated AMP-activated protein kinase α (AMPK α), the energy stress-sensing AMP-activated kinase, and its target Ac-CoA carboxylase (ACC) was increased at an early stage when *Nmrk2* induction began in the *SRF*^{flKO} hearts (Figure 6A and 6B). 5-Aminoimidazole-4-carboxamide ribonucleotide (AICAR) treatment stimulated AMPK phosphorylation in NRCs and increased NMRK2 protein level in NRCs (Figure 6C and 6D). The energetic stress induced by glucose deprivation for 24 hours or FK866 for 48 hours also increased the NMRK2 protein.

AICAR treatment did not modulate NAD^+/NADH but robustly induced *Nmrk2* expression (Figure 6E through 6H). To assess whether the effect of AICAR took place at the level of the *Nmrk2* promoter, we transfected *Nmrk2*-

Figure 4 Continued. group versus NT control cells; ## P <0.01, ### P <0.001 for indicated comparisons. Ad-GFP indicates adenovirus encoding green fluorescent protein; AZ, azaserin; FCCP, Carbonyl cyanide 4-(trifluoromethoxy)phenylhydrazone; FK, FK866; HA, human influenza hemagglutinin tag sequence; NAD, nicotinamide adenine dinucleotide; NRC, neonatal rat cardiomyocyte; and NR, nicotinamide riboside.

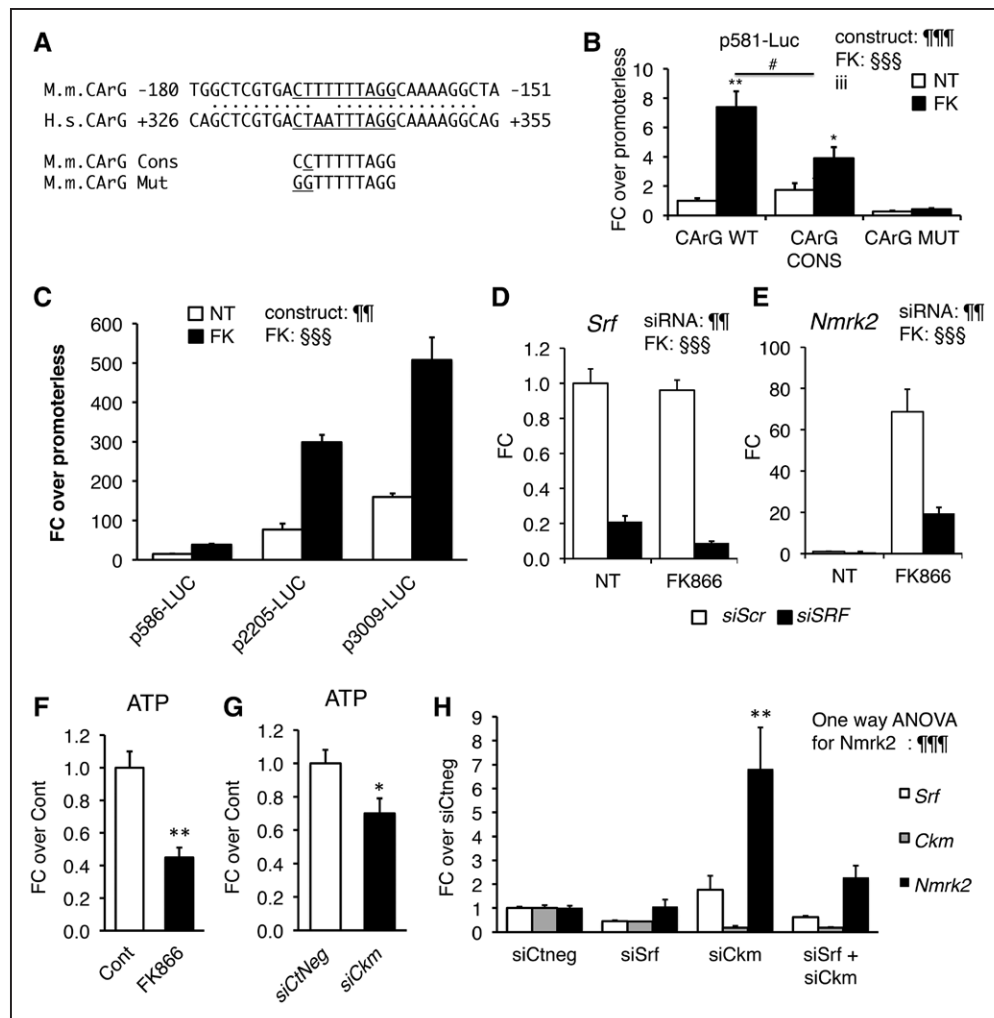


Figure 5. SRF is a component of *Nmrk2* gene transcription.

A, CArG-like binding site for SRF in the conserved region between the murine (M.m) *Nmrk2* promoter and human (H.s) *NMRK2* intron 1. The CArG-like motif was mutated to a CArG consensus sequence (cons) and a CArG mutant (mut) site unable to bind SRF. **B**, Activity of p581-Firefly Luciferase construct bearing wild-type (WT), consensus (CONS), or mutated (MUT) CArG motif without (NT) or with FK866 treatment (10 μ mol/L) in NRC. SV40-renilla luciferase was cotransfected with the *Nmrk2*-Firefly Luc constructs for normalization of transfection efficiency. Two-way factorial ANOVA for independent factors: $***P < 0.001$ for construct effect, $***P < 0.001$ for FK effect; $iii P \leq 0.001$ for the interaction effect. Post hoc Tukey-test: $*P < 0.05$, $**P < 0.01$ FK versus NT; $\#P < 0.05$ CArG construct versus CArG wild type. **C**, Various lengths of the murine *Nmrk2* regulatory region were inserted into pGL4 vector and transfected in nontreated or FK866-treated NRC. Two-way factorial ANOVA for independent factors: $***P < 0.001$ for construct effect, $***P < 0.001$ for FK effect. **D** and **E**, *Srf* and *Nmrk2* mRNA levels in NRC transfected with control scrambled siRNA (siScr) (white bars), or *Srf* or *Nmrk2* siRNA (black bars), without (NT) or with FK866 treatment. Two-way factorial ANOVA for independent factors: $***P < 0.001$ for siRNA effect, $***P < 0.001$ for FK effect. **F** and **G**, ATP levels in NRCs treated for 72 hours with FK866 (**F**) or siCkm (**G**). *t* test: $*P < 0.05$, $**P < 0.01$, between FK-treated cells or siCkm-transfected cells versus control cells or siCtneg-transfected cells, respectively. **H**, *Srf*, *Ckm*, and *Nmrk2* mRNA level in NRC transfected for 72 hours with siRNAs as indicated. A 1-way ANOVA was used. $***P \leq 0.001$. Post hoc Tukey test: $**P < 0.01$, between any group versus siCtneg-transfected control cells. Throughout the figure, data are expressed as mean $\pm$ SEM over control group, fold change over control group (FC) except for **B** and **C**, fold change over promoterless pGL4 plasmid and **H** FC over siCtneg. CArG indicates CC(A/T)₆GG; NRC, neonatal rat cardiomyocyte; siRNA, small interfering RNA; and SRF, Serum Response Factor.

luciferase constructs into NRC and treated the cells with AICAR. The p586-Luc, and moreover, the p3009-Luc reporter, was highly responsive to AICAR treatment (Figure 6I and 6J). Cotransfection of a plasmid overexpressing a dominant negative isoform of AMPK efficiently blunted induction by AICAR (Figure 6I and 6J).

Analysis of the 5'-regulatory sequences of *Nmrk2* revealed enrichment in putative peroxisome proliferator-activated receptor (PPAR) binding sites (Figure 6K). The fragments -3009/-2552 and -1028/-228 contain most of the activity. By comparison, the activity of the *Nmrk2* promoter was very low in cardiac fibroblasts (Figure 6L).

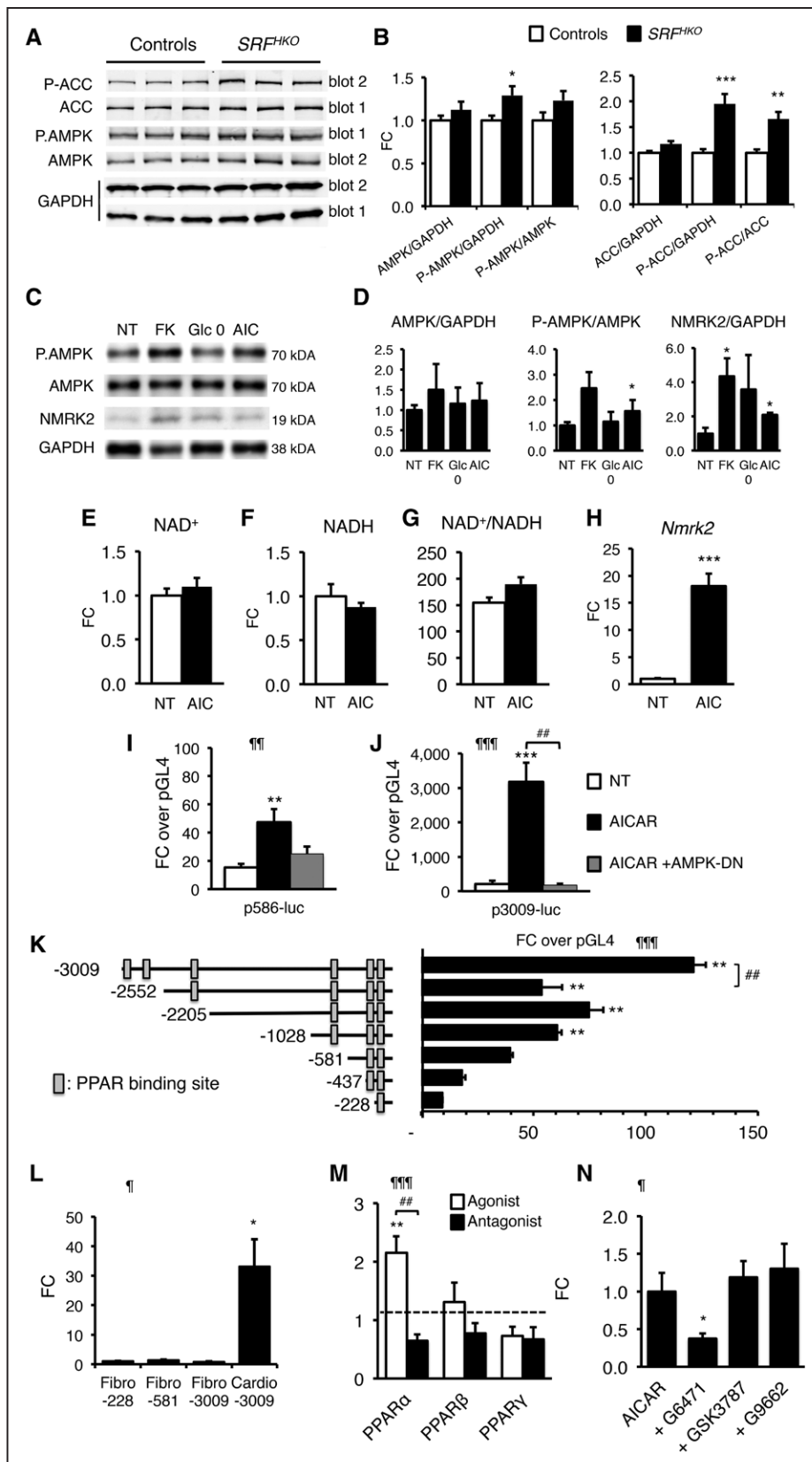


Figure 6. *Nmrk2* expression is increased by AMPK and PPAR α pathways.

A, Representative Western blot analysis of cardiac proteins in control and *SRP^{HKO}* mice at day 9 using antibodies directed against ACC, phospho-ACC (Ser79), AMPK α , and phospho-AMPK α (Thr172). Phosphorylated and total proteins were analyzed on 2 separate gels (gel 1 and 2) and GAPDH antibody was used for loading control. **B**, Quantification (*Continued*)

To assess the role of PPAR factors in *Nmrk2* gene regulation, we cotransfected NRC with the p3009-Luc reporter and PPAR expression vectors. Only PPAR α increased the activity of the *Nmrk2* promoter in the presence of its agonist GW7647 (Figure 6M). The PPAR α antagonist G6471 repressed the *Nmrk2* promoter (Figure 6M) and blunted the activating effect of AICAR (Figure 6N). Neither the PPAR β nor the PPAR γ antagonists had any effect on *Nmrk2* promoter induction by AICAR. These results establish that *Nmrk2* is an AMPK-PPAR α -responsive gene that is induced by energy stress in cardiac cells.

NR Preserves Cardiac Function in the TAC Model

To determine whether NR treatment could be useful in other forms of HF, we assessed its efficiency in the TAC model of pressure overload cardiac hypertrophy. NR was administrated starting 2 days after the TAC or SHAM surgery. Kaplan-Meier survival curve analysis showed no significant difference, mainly because of a similar early mortality between the 2 groups within 1 week after TAC (Figure 7A). NR treatment reduced the final drop in LV ejection fraction that occurred between the fourth and sixth week after TAC (Figure 7B and

Table V in the online-only Data Supplement). NR did not attenuate the dilatation of the LV chamber and had only a transient effect on the thinning of the interventricular septum (Figure 7C and 7D). At euthanization, the cardiac hypertrophy index was similar between the chow diet and NR groups (Figure 7E).

The TAC-induced HF resulted in a decrease of the myocardial NAD⁺ pool, which was partially preserved by NR, with little impact on the NAD⁺/NADH ratio (Figure 7F through 7H). TAC increased *Nmrk2* expression level in the nontreated group and NR-treated group, whereas *Nmrk1* expression remained unchanged and *Nampt* was repressed (Figure 7I through 7K). TAC repressed *Sirt3* expression, *Sirt1* and *Ckm* expression while increasing the expression of *Myh7*, *Anf*, and *Bnp* (Figure 7L through 7Q) but NR treatment had no impact on gene expression.

DISCUSSION

We report that depressed NAD⁺ homeostasis and activation of NMRK2 kinase are early and persistent events in a mouse model of DCM leading to heart failure and that administration of NR, the substrate of NMRK2 and most energetically favorable NAD⁺ precursor, prevents

Figure 6 Continued. of total and phosphoprotein signal from Western blot analyses. Data are normalized on GAPDH signal. The Phos/Total ratio is calculated from GAPDH normalized levels for each individual. n=6 for each group. Data are expressed as mean fold change (FC)±SEM over control group. *t* test: **P*≤0.05, ***P*≤0.01, ****P*≤0.001 over control group. **C**, NRCs were treated with FK866 (10 μmol/L, 72 hours, FK), AICAR (500 μmol/L, 48 hours, AIC), or grown in the absence of glucose in the medium (Glc 0) for 48 hours, and proteins were extracted for Western blot analyses. Representative Western blot. **D**, Quantification on n=3 samples for each condition of NRCs treated as in **C**. Data are expressed as mean FC±SEM over NT group. **P*≤0.05 over control group. **E** through **G**, Intracellular NAD⁺ content (**E**), NADH (**F**), and NAD⁺/NADH ratio (**G**) in NRCs after 24 hours of treatment with AICAR (500 μmol/L). **H**, RT qPCR analysis of *Nmrk2* mRNA level in NRCs treated with AICAR. *t* test: ****P*≤0.001 over nontreated control group. **I** and **J**, NRCs were cotransfected with *Nmrk2*-luciferase constructs containing 586 (**I**) or 3009 base pairs (**J**) of upstream *Nmrk2* regulatory region and a dominant negative (DN) AMPK expression vector. NRCs were transfected at day 3 after plating, followed by AICAR treatment (500 μmol/L) at day 4. Luciferase levels were analyzed at day 5. Normalized Firefly/Renilla values are expressed as in Figure 5B as FC±SEM over the promoterless pGL4 vector. One-way ANOVA: $\eta^2 P \leq 0.01$, $\eta^2 P \leq 0.001$ for treatment effect. Post hoc Tukey test: ***P*≤0.01, ****P*≤0.001 over nontreated control group. ##*P*≤0.01 for AICAR versus AICAR+AMPK-DN. **K**, *Nmrk2* promoter deletion analysis by luciferase assay. Rectangular boxes show the position of the putative PPAR binding sites. Data are expressed as mean FC±SEM over the promoterless pGL4 plasmid. One-way ANOVA: $\eta^2 P \leq 0.001$. Post hoc Tukey test: ***P*≤0.01, over the promoterless pGL4 vector. ##*P*≤0.01 for over the immediately shorter construct. **L**, Neonatal rat cardiac fibroblasts and cardiomyocyte-enriched fractions were separated on a discontinuous Percoll gradient. Cardiac fibroblasts were transfected with the p-228, p-581, and p-3009-FLuc constructs. Cardiomyocytes were transfected with the p-3009-FLuc. SV40-RLuc construct was cotransfected for normalization. Data are expressed as mean FC±SEM over the mean p228-FLuc activity in cardiac fibroblasts. One-way ANOVA: $\eta^2 P \leq 0.05$. Post hoc Tukey test: **P*≤0.05 Cardio-3009 over Fibro-30089. **M**, NRCs were cotransfected with the p3009-FLuc construct and the RXR expression vector and with either PPAR α , PPAR β/δ , or PPAR γ expression vectors. NRCs were treated 24 hours later with the agonists GW7647 (0.6 μmol/L), GW501516 (0.6 μmol/L), and G1929 (0.6 μmol/L), for PPAR α , PPAR β/δ , and PPAR γ , respectively, or with their respective antagonists, GW6471 (10 μmol/L), GSK3787 (2 μmol/L), or GW9662 (2 μmol/L). Data are expressed as mean FC±SEM over normalized luciferase levels of NRCs transfected with the p3009-FLuc construct alone (dashed line) in the same experiment. A 1-way ANOVA was used because each group is independent of the other (different agonists and antagonists): $\eta^2 P \leq 0.001$. Post hoc Tukey test: ***P*≤0.01 versus p3009-FLuc alone. ##*P*≤0.01 for comparison between agonist and antagonist. **N**, NRCs were transfected with the p3009-FLuc construct. Transfected NRCs were treated 24 hours later with the antagonists G6471, GSK3787, or G9662, 30 minutes before adding AICAR for a further 24-hour period. All concentrations were as in **M**. Data are expressed as mean FC±SEM over the p3009-luc construct treated with AICAR alone. One-way ANOVA: $\eta^2 P \leq 0.05$. Post hoc Tukey: **P*≤0.05 versus AICAR treated cells. ACC indicates acetyl-coenzyme A carboxylase; AICAR, 5-aminoimidazole-4-carboxamide ribonucleotide; AMPK, AMP-activated protein kinase; NRC, neonatal rat cardiomyocyte; NT, nontreated cells; PPAR, peroxisome proliferator-activated receptor; and RT qPCR, quantitative reverse transcription polymerase chain reaction.

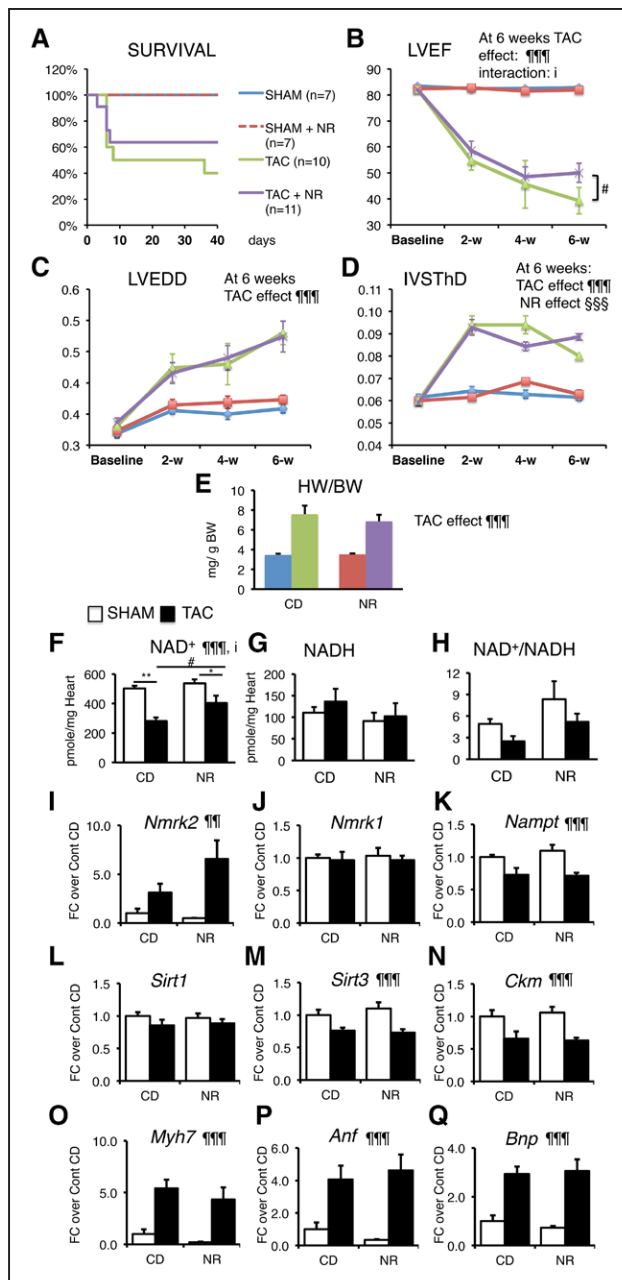


Figure 7. NR treatment preserves myocardial NAD levels and limits the drop in ejection fraction in the pressure overload-induced hypertrophy model.

Two-month-old control and *SRF*^{flKO} male mice were subjected to TAC or SHAM surgery and fed with control chow diet (CD) or NR-enriched diet from day 2 after surgery to day 42. **A**, Kaplan-Meier survival curve analysis. Log-rank statistic: $P>0.05$. Color code shown on the right for each group is valid for **A** through **E**. **B** through **D**, Echocardiography follow-up analysis from baseline (2 days before surgery) to 6 weeks after. **E**, Heart weight to body weight ratio after euthanization at 6 weeks. **F** through **H**, Cardiac NAD⁺ and NADH levels and redox state assessed by the NAD⁺ cycling assay. **I** through **Q**, Cardiac mRNA levels of the indicated genes assessed by RT-qPCR. Data are expressed as mean±SEM. Statistical analysis: 2-way ANOVA for independent factors statistical analysis is shown at (Continued)

deterioration of cardiac function and adverse remodeling (Figure 8).

Altered NAD⁺ homeostasis has been reported in several models of HF including pressure overload, myocardial infarction, and angiotensin 2 infusion.^{3,37,38} In this context, much attention has focused on the NAMPT enzyme that is repressed in several models of cardiac injuries.³⁸ On the other hand, we found a robust upregulation of *Nmrk2* expression in the heart of *SRF*^{flKO} developing DCM and a more modest but significant increase following TAC-induced HF. It is interesting that we observed a similar shift from repressed NAMPT to increased NMRK2 in human failing heart. Interrogation of the Gene Expression Omnibus (GEO) database reveals that *Nmrk2* is induced in several models of cardiomyopathy related to mutations in the *lamin-A* (GEO data set GDS2746),³⁹ *Idh2* (GDS4893),⁴⁰ *PGC1α* (GDS4776),⁴¹ or *Tfr1* (GSE68745)⁴² genes. Considering the relative heterogeneity of molecular functions encoded by these genes, we propose that the activation of the NMRK2 pathway represents a common adaptive mechanism in the failing heart. Here, we show that the *Nmrk2* gene can be activated in response to NAMPT inhibition and activation of the energy stress sensor AMPK. In NRCs, we show that this activation depends on PPARα. PPARα is often expressed at lower levels in the adult failing heart,⁴³ whereas AMPK was shown to activate PPARα in various contexts.⁴⁴ Hence, AMPK activation could compensate in part for the lower levels of PPARα to activate *Nmrk2*. The precise contribution of these actors in the adult failing heart remains an open question. In the case of *SRF*^{flKO} hearts, it is interesting to note that, despite the fact that SRF is an upstream regulator of *Nmrk2* gene that we would expect to be repressed following *Srf* gene inactivation, SRF is also a major upstream regulator of *Ckm* gene, which is important to maintain PCr/ADP/ATP balance,^{15,18} and AMPK is activated early on in the *SRF*^{flKO} heart (this study). We show in vitro that *Ckm* depression alone is sufficient to induce *Nmrk2* expression. In addition *Srf* inactivation alters the expression of numerous sarcomeric genes and triggers a severe eccentric remodeling¹⁵ that is likely to further increase the energetic cost of contraction in vivo. A similar phenomenon could

Figure 7 Continued. 6 weeks and was used for **B** through **Q**, followed by post hoc Tukey test. *** $P\leq 0.01$, **** $P\leq 0.001$ for the TAC effect; **** $P\leq 0.001$ for the NR treatment effect; i $P\leq 0.05$ for the interaction effect. Asterisks indicate statistically significant difference for the indicated comparisons: * $P\leq 0.05$, ** $P\leq 0.01$. # $P\leq 0.05$ for the effect of NR within the TAC group. See accompanying Table V in the online-only Data Supplement for other echocardiography data at 6 weeks and previous stages. IVSThD indicates interventricular septum thickness in diastole; LVEDD, left ventricle end-diastolic diameter; LVEF, left ventricle ejection fraction; NAD, nicotinamide adenine dinucleotide; NR, nicotinamide riboside; RT-qPCR, quantitative reverse transcription polymerase chain reaction; and TAC, transverse aorta constriction.

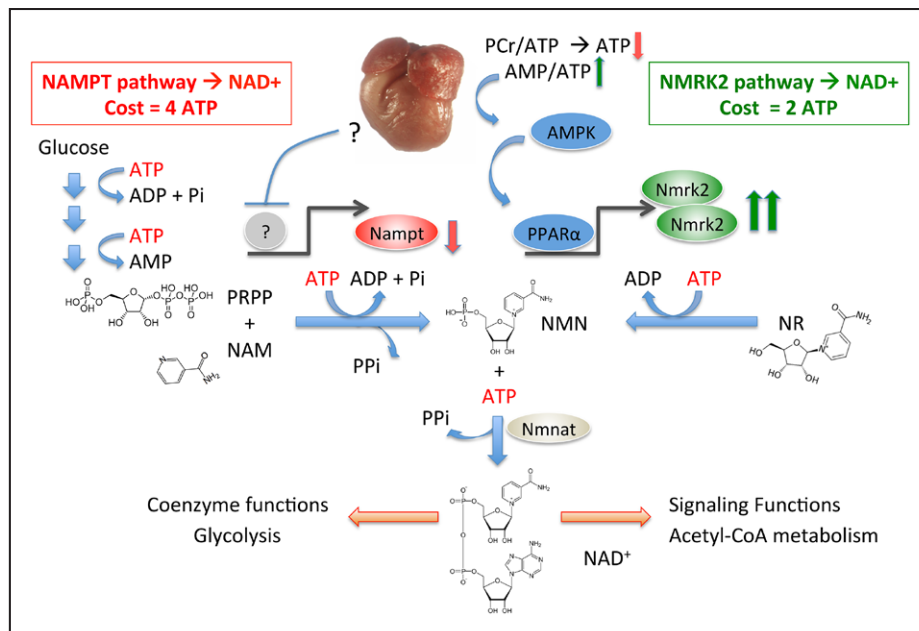


Figure 8. Activation of the NMRK2 pathway to NAD⁺ synthesis as an adaptive energy-sparing mechanism in the failing heart.

DCM is a form of pathological cardiac remodeling that is associated with severe energy depletion leading to HF. The energy stress sensor AMP-kinase is an adaptive signaling pathway aiming to preserve energy in the cells. The NAMPT enzyme is a major, rate-limiting step for NAD⁺ synthesis whose energetic cost equals 4 ATP molecules, including 1 converted to AMP, for the synthesis of 1 NAD⁺. NAMPT is repressed in most forms of HF through an unknown regulatory mechanism. Opposite to this repression, *Nmrk2* gene is an AMPK-responsive gene that is activated in several models of DCM and other forms of pathological remodeling such as the pressure-overload cardiac hypertrophy. NMR-kinase 2 allows the synthesis of NAD⁺ at a lower cost of 2 ATP per molecule. Although NR content may be limited in rodent and human diet, NR can easily be provided as a nutraceutical to help the failing heart to maintain NAD⁺ levels. AMPK indicates AMP-activated protein kinase; DCM, dilated cardiomyopathy; HF, heart failure; NAD, nicotinamide adenine dinucleotide; NAM, nicotinamide; NAMPT, nicotinamide phosphoribosyltransferase; NMN, nicotinamide mononucleotide; NR, nicotinamide riboside; PPAR, peroxisome proliferator-activated receptor; and PRPP, phosphoribosyl pyrophosphate.

occur for instance in the lamin-A mutants, which display severe alterations in cardiomyocyte cytoskeletal integrity and in which SRF activity is repressed and *Nmrk2* expression induced.^{22,39} In addition to its well-known roles in stimulating fuel oxidation to restore ATP levels, we show that AMPK stimulation induces *Nmrk2* expression. NMN synthesis from NR by NMRK enzymes requires a single ATP, whereas synthesis from NAM by NAMPT requires >3 ATP equivalents: one for the autophosphorylation of the enzyme, and 2 (plus a carbohydrate) in formation of phosphoribosyl pyrophosphate (Figure 8). Both pathways consume an additional ATP for fusion with NMN to generate the dinucleotide. Hence, the shift from NAMPT to NMRK2 for NAD⁺ synthesis is an energy-sparing mechanism that may be favored in HF. In turn, we show that one of the beneficial actions of NR, at least in vitro, may be to increase glycolysis in cardiomyocytes. These different mechanisms of action may contribute to increase cardiac efficiency, which should be assessed in future studies to establish whether metabolic substrate preference and potentially glucose oxidation, which is associated with a better respiratory quotient and ATP availability, are improved on NR treatment.

Although the NMRK2 pathway is activated in the failing heart of *SRF^{HKO}* mice, the myocardial NAD⁺ level is depressed, which suggests that circulating and tissue levels of NAD⁺ precursors are insufficient to sustain cardiac NAD⁺ synthesis in mice on a regular rodent diet, stimulating an interest in NR supplementation to correct this defect. A few studies have shown that short-term NAD⁺ supplementation following chronic angiotensin-2 administration³⁷ or NMN supplementation following ischemia-reperfusion injury⁴⁵ or TAC surgery⁴⁶ can normalize alterations in NAD⁺/NADH ratio in the mouse failing heart. It is interesting to note that cells do not take up NMN intact, and, instead, NMN is extracellularly dephosphorylated to NR, such that its cellular conversion to NAD⁺ depends on the NR kinase pathway.^{47,48}

We discovered a strongly beneficial effect of NR in the *SRF^{HKO}* model of HF in the preservation of cardiac function and limitation of cardiac remodeling that was associated with the maintenance of NAD⁺ levels in the heart. NAD⁺ metabolomics and expression analysis showed no evidence for increased levels of PAR, ADPR, Nam, or PARylated proteins, which would indicate a storm of NAD⁺-consuming activities in this model of

HF. Instead, a progressive loss of *Nampt* expression, coupled with the known depression of ATP and high-energy phosphates, is postulated to make it challenging for damaged hearts to maintain NAD⁺ homeostasis.

Beyond the repletion in myocardial NAD⁺ levels, the most striking NR-reversible deficit we found in HF was depression of CS and ACL activity. Many regulatory processes from gene expression to enzyme activity are controlled by reversible protein Lys acetylation. Although there is a variety of acetylases and deacetylases that are NAD⁺-dependent and NAD⁺-independent, production of citrate in mitochondria and its conversion to cytosolic Ac-CoA by the ACL is required to drive changes in substrate fuel selection² and protein acetylation level.^{8,49} NR had no impact on the acetylation level of a mitochondrial protein like Aco2 but robustly increased the acetylation level of FOXO1 or p53 transcription factors in *SRF^{HKO}* hearts and control hearts, suggesting a tight link between NR-derived NAD⁺ synthesis and Ac-CoA availability for posttranslational acetylation of nucleocytoplasmic proteins. This apparently overrides the ability of SIRT1 to deacetylate these factors. At the present stage, it is difficult to assess the consequences of these modifications. Moreover, there is presently no reason to think that FOXO1 and p53 are the only targets whose acetylation level is modified by long-term NR treatment. Rather, our data suggest that the acetylation levels of FOXO1 and p53 can be considered as surrogate markers of an increase in nucleocytoplasmic Ac-CoA metabolism.

Last, we observed that NR led to heightened production of Me4PY, a terminal NAM metabolite associated with local production of H₂O₂. This pathway was shown to promote life extension in the *Caenorhabditis elegans* through a hormetic response to a moderate rise in reactive oxygen species, which activates the cytoprotective NRF-2 transcription factor involved in oxidative stress response.³⁴ In the *SRF^{HKO}* heart, NR was shown to result in Me4PY simultaneous with induction of NRF-2.

Here, we also show a modest although significant protective effect of NR treatment on the drop of LV ejection fraction observed in the TAC model. The effect of NR in the TAC model seems to be less important than in the DCM model, which may be linked to the different nature of the cardiac stress in these models, ie, pressure overload in the TAC model that increases afterload versus volume overload in DCM that increases preload, and the fact that the *Nmrk2* gene is less strongly induced in the TAC model than in the *SRF^{HKO}* model. We note also that the early mortality induced by the TAC led to an early dropout in the untreated and NR-treated TAC groups that may have introduced a bias by removing from the analysis the animals with the worst outcome.

In conclusion, our work shows that oral NR supplementation is a powerful approach to preserve cardiac function and limit remodeling in DCM, a devastating disease currently lacking effective therapies. Recently,

mutations in the de novo NAD⁺ biosynthetic pathway were shown to lead to a new category of genetic diseases called congenital NAD deficiency disorders, which are associated with cardiac and skeletal defects but correctible by niacin supplementation in mouse models of the disease.⁵⁰ Our study and previous findings by others^{13,26–28} suggest that NR has the potential not only to improve the cardiac symptoms in congenital syndromes and acquired HF, but also to act at a systemic level notably on liver metabolism, insulin resistance, and skeletal muscle performances that are altered concomitantly with cardiac defects. However, a limitation of our study is the fact that we did not study the impact of NR on the survival rate of *SRF^{HKO}* mice and the occurrence of terminal congestive HF. Further work will be required to determine the pharmacokinetics and pharmacodynamics of therapeutic NR administration in animal models and people with HF to enable safe and effective human translation.

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Correspondence

Mathias Merckskay, PhD, Signalisation et Physiopathologie Cardiovasculaire, Inserm UMR-S 1180, Faculté de Pharmacie de l'Université Paris-Sud, 5, rue Jean-Baptiste Clément, F-92296 Châtenay-Malabry, France. E-mail mathias.merckskay@inserm.fr

Affiliations

Sorbonne Universités, Université Pierre et Marie Curie Paris 6, Department of Biology of Adaptation and Ageing, CNRS UMR8256, INSERM U1164, Institute of Biology Paris-Seine, DHU FAST, France (N.D., C.T., R.D., A. Gouge, J.B., J.-F.D., Z.L.). Department of Biochemistry, Carver College of Medicine, University of Iowa, Iowa City (S.A.J.T., C.B.). Signalling and Cardiovascular Pathophysiology, UMR-S 1180, Université Paris-Sud, INSERM, Université Paris-Saclay, Châtenay-Malabry, France (C.T., R.D., J.P., M.G., B.M., M.B., A. Garnier, M.M.). Sorbonne Universités, Université Pierre et Marie Curie Paris 6, Plateforme PECMV, UMS28, Paris, France (N.M.). Institute of Metabolism and Systems Research, University of Birmingham, United Kingdom (G.G.L.). Department of Pharmacology and Pharmacotherapy, University of Szeged, Hungary (I.B.). Physiology Department, Faculty of Medicine and EA3072, Université de Strasbourg, France (J.Z.).

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Disclosures

Dr Brenner has an intellectual property interest in uses of NR. He serves on the Scientific Advisory Board of ChromaDex, which distributes NR and is cofounder and the chief scientific advisor of Prohealthspan, which sells NR supplements. The other authors declare no competing financial interest.

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